



Escuela Superior Politécnica del Litoral

Ingeniería de Software II

Taller #2

Continuous Integration

Nombres:

Aragon Intriago Shirley Yamel

Diaz Osorio Fernando Nahim

Muñoz Sanchez Salvador Gabriel

Navarrete Castillo Anthony Josue

Tumbaco Santana Gabriel Alejandro

Fecha:

3 de Junio 2026

Repository URL:

<https://github.com/salva1984/ci-promote-g2>

Features:

1. Register a member, with a name and a unique identifier. - Assigned to Nahim Diaz
2. Loan a book to a member. A book that is already on loan cannot be loaned again. - Assigned to Shirley Aragon
3. Return a book, marking it available again. - Assigned to Gabriel Tumbaco
4. List the books a member currently has on loan. - Assigned to Salvador Muñoz
5. Report the loans that are overdue, given a due date for each loan. - Assigned to Anthony Navarrete

You can find the workflow file in this URL:

<https://github.com/salva1984/ci-promote-g2/blob/main/.github/workflows/ci.yml>

Action tab screenshot (w/passing test)

The screenshot displays the GitHub Actions interface for the repository 'salva1984 / ci-promote-g2'. The 'Actions' tab is selected, showing a workflow run for 'ci.yml' triggered by a pull request. The workflow status is 'Success' with a total duration of 13s. The workflow steps are 'test' (0s) and 'release' (0s). The 'Annotations' section shows a warning about deprecated Node.js 20 actions.

salva1984 / ci-promote-g2

Q Type to search

< Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

feat: Display overdue loans #46

Re-run all jobs

Summary

All jobs

test

release

Run details

Usage

Workflow file

Triggered via pull request 5 minutes ago

lolothens-e opened #10 feature/overload

Status Success

Total duration 13s

Artifacts -

ci.yml

on: pull_request

test 0s

release 0s

Annotations

1 warning

test

Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, actions/setup-p...

Show more

Branch protection settings screenshot:

☒ **Require a pull request before merging**
Require all commits be made to a non-target branch and submitted via a pull request before they can be merged.

Hide additional settings ^

Required approvals

1 ▾

The number of approving reviews that are required before a pull request can be merged.

☐ **Dismiss stale pull request approvals when new commits are pushed**
New, reviewable commits pushed will dismiss previous pull request review approvals.

☐ **Require review from specific teams**
A collection of reviewers and associated file patterns. Each reviewer has a list of file patterns which determine the files that reviewer is required to review.

☐ **Require review from Code Owners**
Require an approving review in pull requests that modify files that have a designated code owner.

☐ **Require approval of the most recent reviewable push**
Whether the most recent reviewable push must be approved by someone other than the person who pushed it.

☐ **Require conversation resolution before merging**
All conversations on code must be resolved before a pull request can be merged.

Allowed merge methods

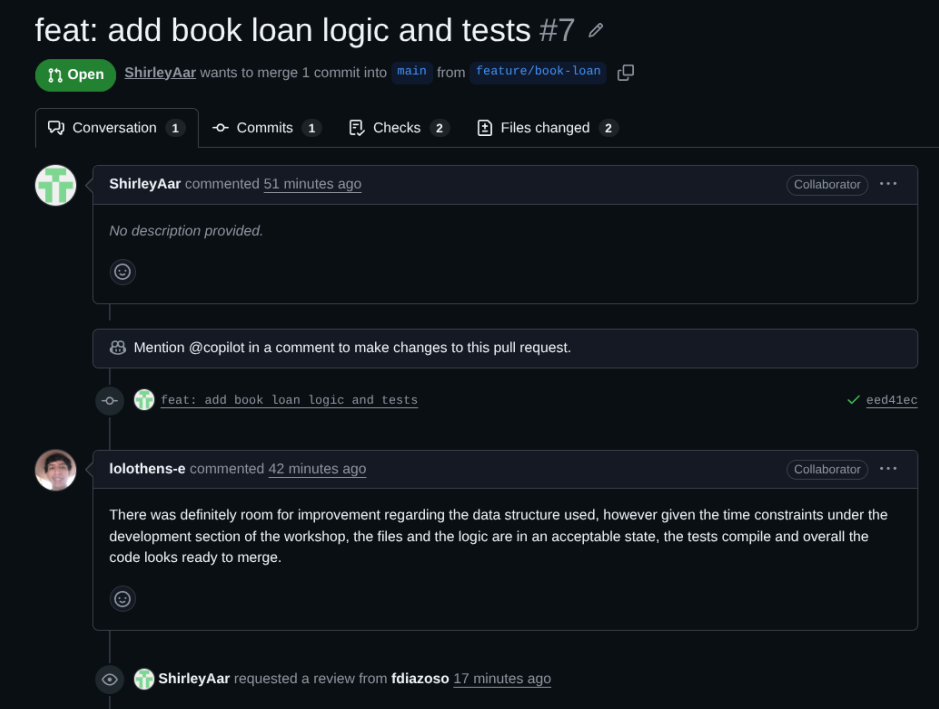
Merge, Squash, Rebase ▾

When merging pull requests, you can allow any combination of merge commits, squashing, or rebasing. At least one option must be enabled.

Pull requests opened and reviewed:

Aragon Intriago Shirley Yamel

Opened: <https://github.com/salva1984/ci-promote-g2/pull/7>



feat: add book loan logic and tests #7

ShirleyAar wants to merge 1 commit into main from feature/book-loan

Conversation 1 | Commits 1 | Checks 2 | Files changed 2

ShirleyAar commented 51 minutes ago

No description provided.

Mention @copilot in a comment to make changes to this pull request.

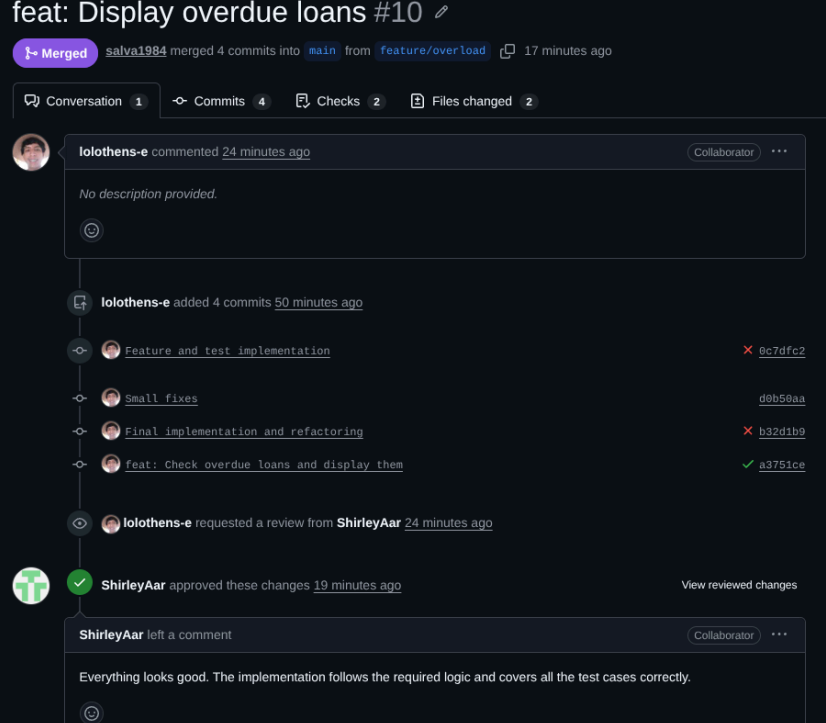
feat: add book loan logic and tests ✓ eed41ec

lolothens-e commented 42 minutes ago

There was definitely room for improvement regarding the data structure used, however given the time constraints under the development section of the workshop, the files and the logic are in an acceptable state, the tests compile and overall the code looks ready to merge.

ShirleyAar requested a review from fdiazoso 17 minutes ago

Reviewed: <https://github.com/salva1984/ci-promote-g2/pull/10>



feat: Display overdue loans #10

salva1984 merged 4 commits into main from feature/overload 17 minutes ago

Conversation 1 | Commits 4 | Checks 2 | Files changed 2

lolothens-e commented 24 minutes ago

No description provided.

lolothens-e added 4 commits 50 minutes ago

- Feature and test implementation ✗ 0c7dfc2
- Small fixes d9b56aa
- Final implementation and refactoring ✗ b32d1b9
- feat: Check overdue loans and display them ✓ a3751ce

lolothens-e requested a review from ShirleyAar 24 minutes ago

ShirleyAar approved these changes 19 minutes ago

ShirleyAar left a comment

Everything looks good. The implementation follows the required logic and covers all the test cases correctly.

Diaz Osorio Fernando Nahim

Opened: <https://github.com/salva1984/ci-promote-g2/pull/4>

ADD FEATURE: register member and test for the new feature #4

Merged Gatumbac merged 1 commit into main from feature/register_member 2 hours ago

Conversation 1 Commits 1 Checks 2 Files changed 2

fdiazoso commented 2 hours ago Collaborator ...

Add a feature for add members

ADD FEATURE: register member and test for the new feature ✓ f65e331

Gatumbac approved these changes 2 hours ago View reviewed changes

Gatumbac approved these changes 2 hours ago View reviewed changes

Gatumbac left a comment • edited Collaborator ...

All changes looks good, no comments and no code smells, short and well scoped functions. GJ Nahim!

Gatumbac merged commit 6106da7 into main 2 hours ago 2 checks passed View details Revert

Reviewed: <https://github.com/salva1984/ci-promote-g2/pull/3>

chore: drop comments and correct spaces #3

Merged fdiazoso merged 1 commit into main from chore/drop-comments 3 hours ago

Conversation 1 Commits 1 Checks 2 Files changed 2

Gatumbac commented 3 hours ago Collaborator ...

No description provided.

chore: drop comments and correct spaces ✓ 60be238

Gatumbac requested a review from salva1984 3 hours ago

fdiazoso self-assigned this 3 hours ago

fdiazoso approved these changes 3 hours ago View reviewed changes

fdiazoso left a comment Collaborator ...

lgm

fdiazoso merged commit 608d376 into main 3 hours ago 2 checks passed View details Revert

Muñoz Sanchez Salvador Gabriel

Opened: <https://github.com/salva1984/ci-promote-g2/pull/6>

This screenshot shows a GitHub Pull Request titled "feat: List the books a member currently has on loan. #6". The pull request is merged, with the message "lolothen-e merged 1 commit into main from feature/list-book-ownership 22 minutes ago". The interface includes tabs for Conversation (1), Commits (1), Checks (2), and Files changed (2). A comment from the owner, salva1984, states "No description provided." Below this, a commit history shows the pull request's origin. A review from lolothen-e is shown, with a green checkmark indicating approval 22 minutes ago. The reviewer's comment reads: "The code looks good, the tests are passing with flying colors, and the logic remains intact so far".

Reviewed: <https://github.com/salva1984/ci-promote-g2/pull/5>

This screenshot shows a GitHub Pull Request titled "Feature/return book #5". The pull request is merged, with the message "salva1984 merged 3 commits into main from feature/return-book 2 hours ago". The interface includes tabs for Conversation (1), Commits (3), Checks (2), and Files changed (5). A comment from the collaborator, Gatumbac, states "No description provided." Below this, a commit history shows the pull request's origin. A review from salva1984 is shown, with a green checkmark indicating approval 2 hours ago. The reviewer's comment reads: "clean, modular code free of code smells and sufficient test coverage in the new code, LGTM".

Navarrete Castillo Anthony Josue

Opened: <https://github.com/salva1984/ci-promote-g2/pull/10>

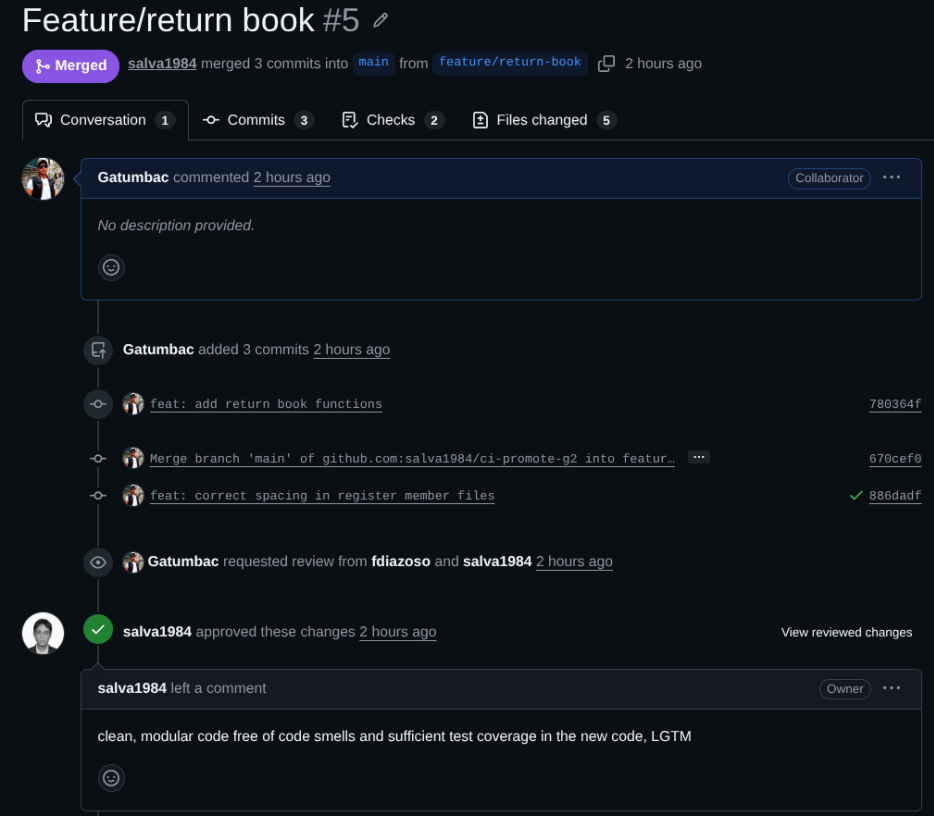
The screenshot shows a GitHub pull request interface. At the top, the title is "feat: Display overdue loans #10". Below the title, it says "lolothen-e wants to merge 4 commits into main from feature/overload". The status is "Awaiting approval". The pull request has 0 conversations, 4 commits, 2 checks, and 2 files changed. A comment from lolothen-e says "No description provided." and "Mention @copilot in a comment to make changes to this pull request." The commit history shows: "Feature and test implementation" (0c7dfc2), "Small fixes" (d0b50aa), "Final implementation and refactoring" (b32d1b9), and "feat: Check overdue loans and display them" (a3751ce). A "Review required" message states: "At least 1 approving review is required by reviewers with write access." The "All checks have passed" message says: "1 skipped, 1 successful checks". The right sidebar shows the "Reviewers" section with ShirleyAar and Copilot. The "Assignees" section says "No one—assign yourself". The "Labels" section says "None yet". The "Projects" section says "None yet". The "Milestone" section says "No milestone". The "Development" section says "Successfully merging this pull request may close these issues." and "None yet". The "Notifications" section says "Unsubscribe" and "You're receiving notifications because you authored the".

Reviewed: <https://github.com/salva1984/ci-promote-g2/pull/6>

The screenshot shows a GitHub pull request interface. At the top, the title is "feat: List the books a member currently has on loan. #6". Below the title, it says "lolothen-e merged 1 commit into main from feature/list-book-ownership". The status is "Merged". The pull request has 1 conversation, 1 commit, 2 checks, and 2 files changed. A comment from salva1984 says "No description provided." and "feat: List the books a member currently has on loan." (eabbb0f). A comment from lolothen-e says "The code looks good, the tests are passing with flying colors, and the logic remains intact so far". The commit history shows: "feat: List the books a member currently has on loan." (eabbb0f). A "Pull request successfully merged and closed" message says: "You're all set — the feature/list-book-ownership branch can be safely deleted." and "Delete branch". The right sidebar shows the "Reviewers" section with lolothen-e (checked), ShirleyAar, and Copilot. The "Assignees" section says "No one—assign yourself". The "Labels" section says "None yet". The "Projects" section says "None yet". The "Milestone" section says "No milestone". The "Development" section says "Successfully merging this pull request may close these issues." and "None yet". The "Notifications" section says "Unsubscribe" and "You're receiving notifications because you modified the open/close state.".

Tumbaco Santana Gabriel Alejandro

Opened: <https://github.com/salva1984/ci-promote-g2/pull/5>



Feature/return book #5

Merged salva1984 merged 3 commits into `main` from `feature/return-book` 2 hours ago

Conversation 1 | Commits 3 | Checks 2 | Files changed 5

Gatumbac commented 2 hours ago Collaborator

No description provided.

Gatumbac added 3 commits 2 hours ago

- feat: add return book functions 780364f
- Merge branch 'main' of github.com:salva1984/ci-promote-g2 into featur... 670cef0
- feat: correct spacing in register member files 886dadf

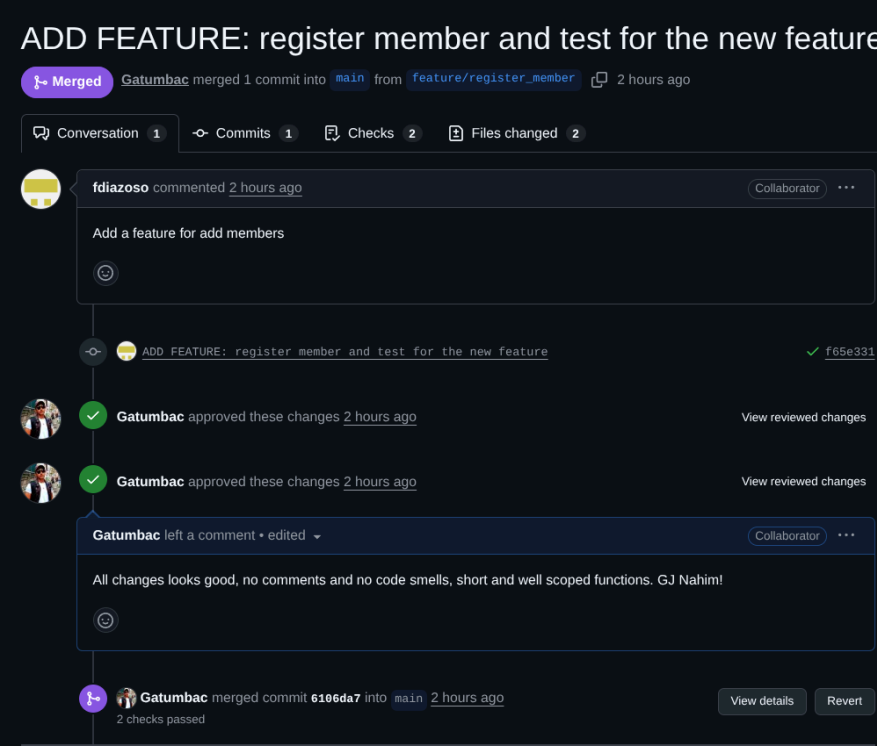
Gatumbac requested review from **fdiazoso** and **salva1984** 2 hours ago

salva1984 approved these changes 2 hours ago View reviewed changes

salva1984 left a comment Owner

clean, modular code free of code smells and sufficient test coverage in the new code, LGTM

Reviewed: <https://github.com/salva1984/ci-promote-g2/pull/4>



ADD FEATURE: register member and test for the new feature

Merged Gatumbac merged 1 commit into `main` from `feature/register_member` 2 hours ago

Conversation 1 | Commits 1 | Checks 2 | Files changed 2

fdiazoso commented 2 hours ago Collaborator

Add a feature for add members

ADD FEATURE: register member and test for the new feature f65e331

Gatumbac approved these changes 2 hours ago View reviewed changes

Gatumbac approved these changes 2 hours ago View reviewed changes

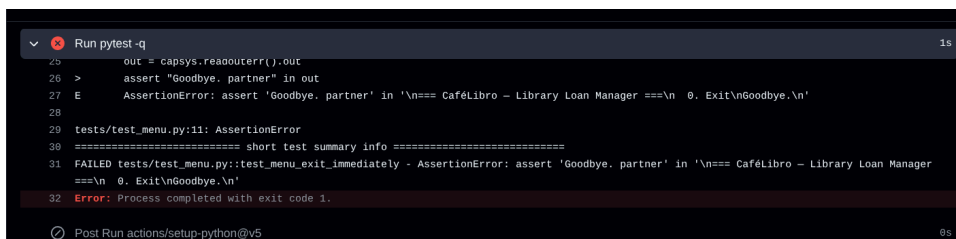
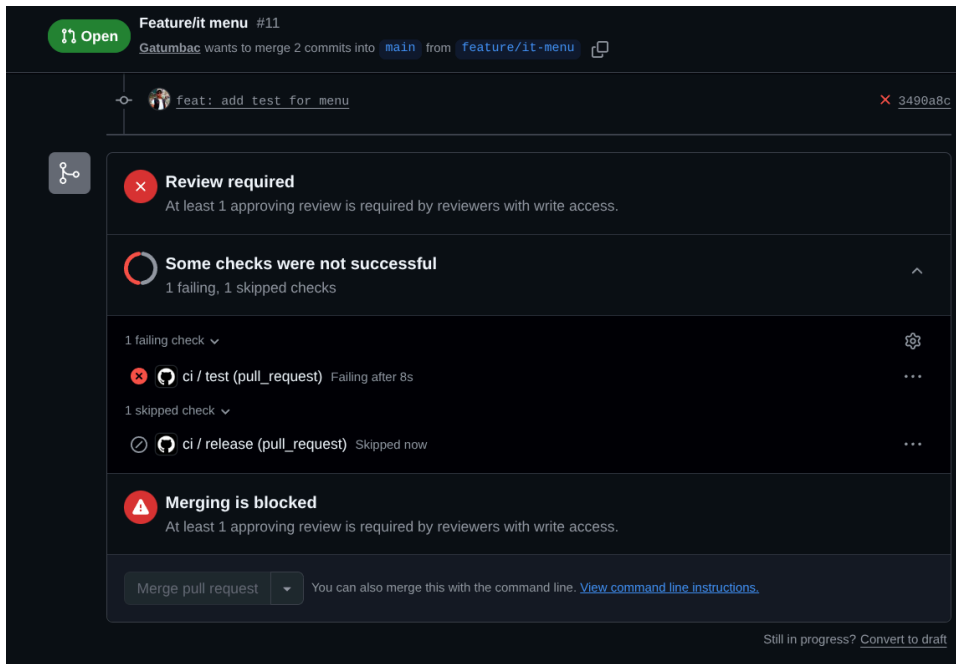
Gatumbac left a comment • edited Collaborator

All changes looks good, no comments and no code smells, short and well scoped functions. GJ Nahim!

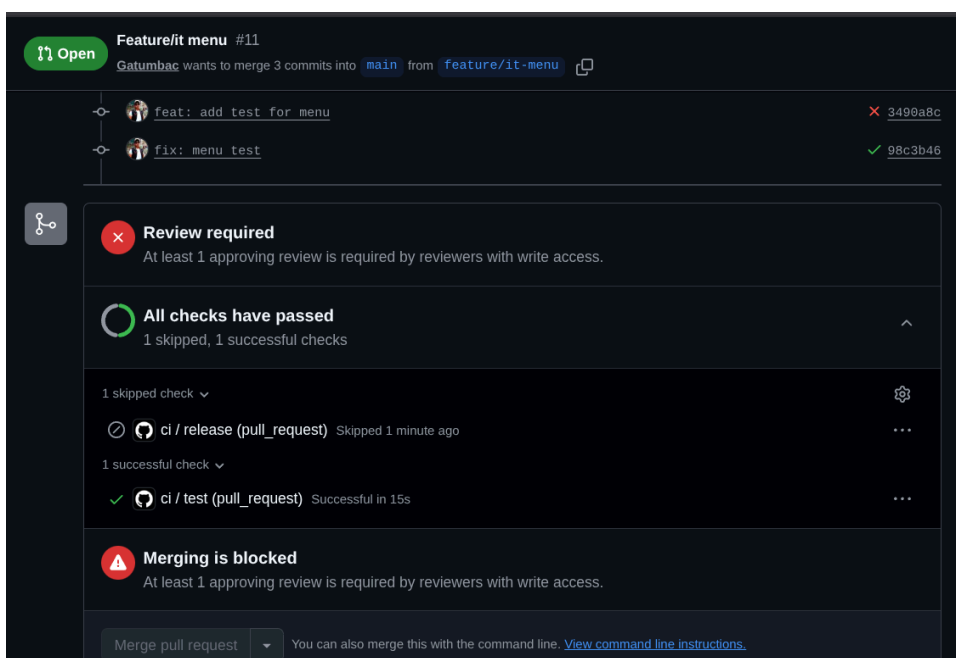
Gatumbac merged commit 6106da7 into `main` 2 hours ago 2 checks passed View details Revert

Pull request:

Failing screenshot:



Passing screenshot:



Release automation:

```
ci-promote-g2 / .github / workflows / ci.yml
Code Blame 52 lines (47 loc) · 1.34 KB
21     release:
22         needs: test
23         if: github.event_name == 'push' && github.ref == 'refs/heads/main'
24         runs-on: ubuntu-latest
25         permissions:
26             contents: write
27         steps:
28             - uses: actions/checkout@v4
29               with:
30                 fetch-depth: 0
31             - uses: actions/setup-python@v5
32               with:
33                 python-version: "3.12"
34             - name: Install build tool
35               run: pip install build
36             - name: Build package artifact
37               run: python -m build
38             - name: Upload packaged artifact
39               uses: actions/upload-artifact@v4
40               with:
41                 name: cafelibro-packaged-distribution
42                 path: dist/
43             - name: Create version tag
44               run: |
45                 git config --global user.name "github-actions[bot]"
46                 git config --global user.email "github-actions[bot]@users.norep
47
48                 TAG_VERSION="v1.0.${{ github.run_number }}"
49                 echo "Creando tag de versión: $TAG_VERSION"
50
51                 git tag $TAG_VERSION
52                 git push origin $TAG_VERSION
```

```
release
succeeded 35 minutes ago in 9s
Q Search logs
🔄 ⚙️
📁 Build package artifact
1s
📁 Upload packaged artifact
1s
1 ▶ Run actions/upload-artifact@v4
16 With the provided path, there will be 2 files uploaded
17 Artifact name is valid!
18 Root directory input is valid!
19 Beginning upload of artifact content to blob storage
20 Uploaded bytes 6815
21 Finished uploading artifact content to blob storage!
22 SHA256 digest of uploaded artifact zip is 586632eca055eef621abe23cbe52f2126834e4cea67c310b5cad25ea071680c3
23 Finalizing artifact upload
24 Artifact cafelibro-packaged-distribution.zip successfully finalized. Artifact ID 7401573579
25 Artifact cafelibro-packaged-distribution has been successfully uploaded! Final size is 6815 bytes. Artifact ID is 7401573579
26 Artifact download URL: https://github.com/salva1984/ci-promote-g2/actions/runs/26926343503/artifacts/7401573579
1s
📁 Create version tag
1s
1 ▶ Run git config --global user.name "github-actions[bot]"
18 Creando tag de versión: v1.0.47
19 To https://github.com/salva1984/ci-promote-g2
20 * [new tag] v1.0.47 -> v1.0.47
```



Continuous Integration decision:

1. To switch from CI to CD, branch protection rules must be updated to allow automated auto-merging instead of requiring manual peer reviews. Additionally, a deployment step must be added to the pipeline to automatically publish or distribute the compiled application as soon as the tests pass.
2. The most dangerous feature to automate is "Loan a book" because it handles critical validation limits. Since the application relies on a single shared JSON file to persist data, an undetected bug could easily corrupt or overwrite the file, causing permanent data loss and disrupting library operations.
3. To mitigate the inherent dangers of Continuous Deployment, advanced engineering teams implement automated rollbacks as a crucial safety net before trusting automatic releases. This mechanism links the deployment pipeline with real-time monitoring tools (such as Prometheus or Grafana) that track strict system health thresholds, such as a command error rate exceeding 5% or unhandled database exceptions. If a newly deployed automated release breaches these thresholds during its initial minutes in production, the monitoring system autonomously triggers an alert that executes a roll-back script in the pipeline
Cited from : <https://ctox.com/best-practices-for-rollback-plan-in-secure-deployments/>
4. As a development team, we have decided not to enable Continuous Deployment for CaféLibro at this time. The application is in an early development phase structured around a short-duration workshop, meaning our current automated pytest suite is basic and lacks the coverage required to catch complex concurrency flaws or structural JSON corruption. Handing off direct-to-production deployment relying solely on

minimal unit tests would be highly irresponsible for the library's business stability. For us to reconsider this stance in the future, several technical milestones must be achieved: elevating code coverage above 95% with robust integration tests for JSON persistence, establishing an automated staging environment to simulate real terminal CLI usage, and deploying an automated backup system to clone the JSON state before any live software update.

Questions

1. Why is continuous integration worth the setup effort on a team project? What goes wrong on a shared repository without it?

Setting up CI ensures that code conflicts and system failures are detected immediately after they are introduced. Because developers integrate in small increments, pinpointing the exact cause of a broken test is straightforward, which keeps the cost and time to fix it low while maintaining a main branch that is always correct and passing. Without CI, developers could work in isolation for extended periods and integrate their changes late. This results in a lot of merge conflicts, the resulting bugs are incredibly difficult to locate and resolve.

2. This workshop used GitHub Actions. Research at least two other tools that do continuous integration and delivery (for example Jenkins, GitLab CI, CircleCI, Travis CI, or others). For each, say briefly how a project configures it and where it runs the build.

Jenkins:

Configuration: Set up using a domain-specific language (Groovy syntax) inside a text file named a Jenkinsfile, or configured manually via its web interface.

Build Environment: Runs the build on dedicated integration servers, cloud virtual machines, or self-hosted local infrastructure managed entirely by the user.

GitLab CI:

Configuration: Configured using a structured .gitlab-ci.yml file placed directly in the root directory of the repository.

Build Environment: Executes builds on "GitLab Runners", which can be hosted in GitLab's cloud ecosystem or installed on a team's own private server hardware.

3. Of the tools you compared, including GitHub Actions, which would you choose for the course project, and why? Consider where your code lives, the cost, and how much setup each one needs.

We choose GitHub Actions for the following reasons:

- **Code Location:** The team's public repository is already hosted natively on GitHub, completely eliminating the need to sync external platforms or handle third-party webhooks.
- **Cost:** It is entirely free of charge for public open-source repositories, removing any financial barriers for a student budget.
- **Setup Effort:** The setup overhead is virtually zero. There are no servers to maintain or software packages to install; configuration requires nothing more than adding a ci.yml file under the .github/workflows/ directory.

4. How does continuous integration fit with the way agile teams work in short iterations?

Agile teams operate in rapid, short iterations designed to deliver functional software continuously. Continuous Integration acts as the technical backbone for this rhythm by allowing developers to safely merge small increments of code frequently. Automated testing on a dedicated server acts as an immediate safety net, ensuring that fast-paced changes do not cause regressions or break existing features. This keeps the codebase perpetually stable and deployable at the end of every sprint cycle.

Conclusions

- Automating tests with pytest in GitHub Actions within the ci-promote-g2 repository ensured the early detection of bugs before each integration. This, combined with branch protection rules requiring peer reviews and mandatory passing checks, effectively shielded the stability of the main branch.
- The pipeline practically demonstrated its value as a quality guardrail when an intentional breaking commit in the menu component triggered a test failure and successfully blocked the merge. Once the assertion error was corrected, the pipeline seamlessly flipped to green and unblocked the integration, proving that automated guardrails effectively isolate defects on developer branches before they can degrade the mainline's integrity.

Recommendations

- It is highly recommended to replace the flat JSON file with a relational database to handle concurrency safely and eliminate data corruption risks. Additionally, the unit and integration test suites must expand to exceed 95% code coverage before the team considers any future transition to CD
- Setting up an automated staging environment to accurately simulate the terminal CLI's real-world behavior is highly advisable. Furthermore, an automated backup system should be integrated to clone the data state immediately prior to any software deployment in production.

